

Devran V. Turson

dvturson@gmail.com | [\(571\) 365-6107](tel:(571)365-6107) | Centreville, VA | github.com/dvturson | linkedin.com/in/devran-turson/ | dvturson.github.io/

EDUCATION

James Madison University	BS in Computer Science Harrisonburg, VA • May 2028
--------------------------	---

EXPERIENCE

IMADE3D	Centreville, VA
---------	-----------------

Independent Contractor – Embedded Systems Software May 2026 – July 2026

- Developed embedded firmware APIs for the BBC micro:bit V2 (ARM Cortex-M0), implementing hardware drivers for an SSD1306 OLED display (I²C), NeoPixel LEDs, ultrasonic sensors, and servo motor control
- Engineered a custom SSD1306 framebuffer renderer with direct I²C peripheral control, bypassing driver limitations to enable pixel-accurate filled-shape rendering on a resource-constrained microcontroller
- Built a hardware documentation repository covering verified components, wiring guides, and ready-to-flash .hex firmware binaries for classroom deployment

James Madison University	Harrisonburg, VA
Haynes Leader / Teaching Assistant	October 2025 – Present

- Represent the Haynes program at Open House and CHOICES recruitment events; follow up with prospective students
- Create and maintain contact lists for current and former Haynes scholars to support program communications
- Serve as a Learning Assistant for Math 167 (Haynes Seminar), supporting student learning during Monday/Wednesday sessions
- Onboard incoming Haynes Scholars via email, coordinating course registration, housing, and faculty advisor details
- Coordinate social events and communications across the Haynes community

JEM Lab Technician	March 2025 – Present
--------------------	----------------------

- Operated and maintained FDM and resin 3D printers (Ultimaker S3, Form 3+) using Cura, Preform, and OpenSCAD; designed STL files and handled routine troubleshooting
- Produced 3D printed mathematical models for STEM outreach and educational demonstrations

First Year Student Researcher – FYRE Program (Twoville)	January 2025 – May 2025
---	-------------------------

- Collaborated on Twoville, a bidirectional code-and-mouse programming language for 2D/3D digital fabrication

PROJECTS

Routine Clicker – Full Stack	July 2026
------------------------------	-----------

- Designed and built a custom ESP32-based hardware device integrating a physical button, OLED display, and Wi-Fi module for real-time habit tracking
- Developed a JavaScript/HTML/CSS web dashboard featuring a GitHub style heatmap calendar and OLED progress bar for commit tracking
- Implemented C++ firmware on the ESP32 microcontroller managing Wi-Fi stack initialization, HTTPS REST communication, and real-time data synchronization within a resource-constrained embedded environment

pxt-jellysystem – MakeCode Extension	May 2026
--------------------------------------	----------

- Built a MakeCode/TypeScript extension for the BBC micro:bit V2 (ARM Cortex-M0) enabling hardware control of OLED displays, NeoPixel LEDs, servo motors, and ultrasonic sensors through a student-friendly block interface
- Designed a unified hardware abstraction layer managing I²C and PWM peripheral configurations, allowing servo setups to be swapped without changes to student-facing code
- Restructured the extension's block menu for student readability, matching group names to physical hardware labels for intuitive navigation

SKILLS

Languages	Tools
C++, Python, Java, C, TypeScript, JavaScript, HTML, CSS, SQL, OpenSCAD, Markdown, NetLogo, Twoville	Git, GitHub, VS Code, Linux/Unix, Node.js, REST APIs, Firebase, Arduino IDE, BBC micro:bit, MakeCode, GCC, GNU Make, Figma, Cura, Preform